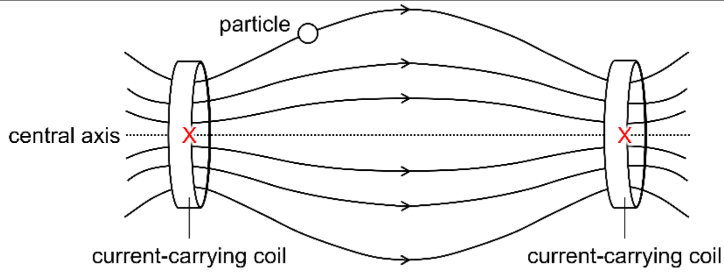
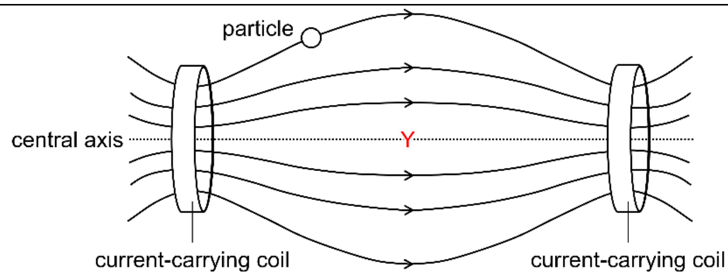
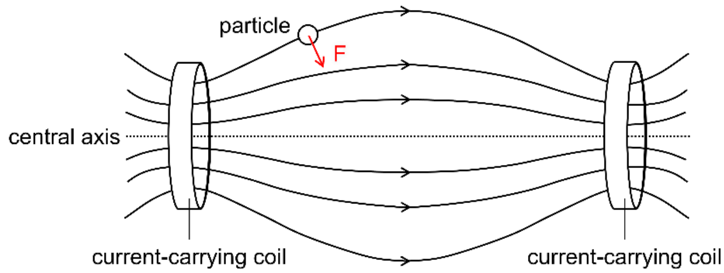
8(a)	$E = 28.368 - 2.230 - 8.499$ $= 17.64$
	$E \approx 17.6$ MeV
(b)	The ‘Coulomb barrier’ is the electric potential energy due to the electrostatic repulsion between the positively charged nuclei that must be overcome for nuclear fusion to occur.
(c)(i)1.	
(i)2.	
(i)3.	
(c)(ii)	The nucleus will not have a component of velocity that is perpendicular to the magnetic field. Hence, there will be no magnetic force acting on the nucleus.
(d)(i)	Changing the current through the central solenoid changes the magnetic field produced by the central solenoid.
	This changes the magnetic flux linkage in the plasma. By Faraday’s law, an e.m.f. is induced in the plasma.
	This causes the charged particles in the plasma to flow around the torus producing plasma current.
(d)(ii)	the resistivity of the plasma is $1.8 \times 10^{-9}\ \Omega$ m
	$R = \dfrac{\rho l}{A}$ $= \dfrac{\rho(2\pi R)}{\pi a^2}$ $= \dfrac{(1.8 \times 10^{-9})(2\pi)(6.2)}{\pi(2.0)^2}$ $= 5.58 \times 10^{-9}$
	$R \approx 5.6 \times 10^{-9}\ \Omega$

(d)(iii)	$P = I^2R$ $= \left(15\times10^6\right)^2\left(5.6\times10^{-9}\right)$ $= 1.26\times10^6$ $\approx 1.3\times10^6\,\text{W}$
(e)	Magnetic force provides for the centripetal force.
	$Bqv = mr\omega^2$ $= m\left(r\omega\right)\omega$ $= mv\omega$ $\omega = \dfrac{Bq}{m}$ $= \dfrac{(5.3)\left(1.60\times10^{-19}\right)}{2\times1.66\times10^{-27}}$ $= 2.554\times10^8\ \text{rad s}^{-1}$
	$2\pi f = 2.554\times10^8$ $f = \dfrac{2.554\times10^8}{2\pi}$ $= 40.65\times10^6\ \text{Hz}$ $\approx 41\,\text{MHz}$
(f)	Neutral beams are unaffected by the magnetic fields hence they are able to propagate in a straight line before entering the into plasma.

(g)	volume of the torus $V = 2\pi^2 R a^2$ $= 2\pi^2 (6.2)(2^2)$ $= 489.53 \text{ m}^3$
	$S = kn^2 E$ $= (3.57 \times 10^{-23})(10^{20})^2 (17.6 \times 10^6 \times 1.60 \times 10^{-19})$ $= 1.0053 \times 10^6 \text{ W m}^{-3}$
	$Q = \dfrac{SV}{P_{\text{CRH}} + P_{\text{NBI}}}$ $= \dfrac{(1.0053)(489.53)}{20 + 33}$ $= 9.29$ ≈ 9.3